

Smart Traffic Light Controller using RL

Step-by-Step Installation & Execution Manual for Any Computer

Repository: github.com/SauravLamba17/RL-Smart-Traffic-Light	Author: Saurav Lamba
Environment: Python 3.10 – 3.14 (Cross-Platform)	Core Stack: Q-Learning, Pygame-CE, Streamlit

1. Project Overview & Capabilities

This project replaces rigid fixed-timer intersection lights with an autonomous **Tabular Q-Learning** reinforcement learning agent. The agent observes 4-lane queue sizes, detects emergency vehicle presence (Tier 1 Fire Trucks & Tier 2 Ambulances), and optimizes signal timing in real time. The repository comes fully bundled with a pre-trained model, animated Pygame desktop simulation, and a rich 5-page Streamlit analytical web dashboard.

2. System Prerequisites

Ensure the target machine meets the following basic requirements:

- **Operating System:** Windows 10/11, macOS (Intel or Apple Silicon), or Linux (Ubuntu/Debian/Fedora).
- **Python:** Python 3.10 to Python 3.14. (*Ensure 'Add python.exe to PATH' was checked during setup*).
- **Hardware:** Any dual-core CPU with at least 4 GB RAM. **No GPU required** (< 0.1 ms lookup per frame).
- **Git:** Installed for cloning (or simply use direct ZIP download).

3. Step-by-Step Setup Instructions

Step 3.1: Download or Clone the Repository

Open a Terminal (macOS/Linux) or PowerShell / Command Prompt (Windows) and run:

```
git clone https://github.com/SauravLamba17/RL-Smart-Traffic-Light.git
cd RL-Smart-Traffic-Light
```

*Alternative without Git: Visit [GitHub](#) → green **Code** button → **Download ZIP**, and extract.*

Step 3.2: Create and Activate a Virtual Environment (Recommended)

Creating a virtual environment ensures zero library conflicts with existing system packages:

```
Windows (PowerShell or Command Prompt):
python -m venv venv
.\venv\Scripts\activate
macOS / Linux (Bash or Zsh):
python3 -m venv venv
source venv/bin/activate
```

Step 3.3: Install Dependencies

Install all runtime dependencies with a single pip command:

```
pip install -r requirements.txt
```

Packages: pygame-ce, streamlit, pandas, plotly, matplotlib, numpy.

4. How to Run the Software

The project is engineered for zero-friction launch. You can run the entire system with one single command, or launch individual components independently.

Method A: The Unified Launcher (Recommended)

Run the all-in-one execution controller:

```
python run.py
```

On Windows, you can also simply double-click **START_SIMULATION.bat**.

What happens automatically:

- Model Verification:** Checks models/trained_q_table.pkl (bundled in repo, ready out-of-the-box).
- Web Dashboard:** Starts Streamlit on port 8501 and opens browser to <http://localhost:8501>.
- Interactive Desktop Simulation:** Launches the animated Pygame launch screen. Use $\uparrow/\downarrow$ and press **ENTER**.
- Clean Shutdown:** Closing the Pygame window automatically halts all background processes.

Method B: Launching Standalone Components

Target Action	Command Line Execution	Description
Live Simulation	python main.py --simulate	Directly opens animated Pygame window.
Web Analytics	streamlit run streamlit_app/app.py	Opens the 5-page analytical dashboard.
Retrain Agent	python main.py --train	Trains model from scratch (2500 ep, ~2.5 min).
Benchmark Eval	python main.py --eval	Runs 50-episode headless statistical benchmark.
Generate Plots	python main.py --plot	Exports high-res PNG plots into analytics/plots/.

5. Interactive Pygame Simulation Controls

While the live side-by-side simulation is active, interact using these keys:

Key Binding	Functionality & Demonstration Highlight
1, 2, 3, 4	Dispatches an Ambulance (Tier 2 Priority) into North, South, East, or West lane.
5, 6, 7, 8	Dispatches a Fire Truck (Tier 1 Critical) into North, South, East, or West lane.
F or C	Emergency Conflict Test: Simultaneously spawns Fire Truck (North) and Ambulance (East) to demonstrate Tier 1 vs Tier 2 priority arbitration!
SPACE	Pauses / Resumes the simulation physics.
UP / DOWN	Cycles simulation speed: 1x $\rightarrow$ 2x $\rightarrow$ 4x $\rightarrow$ 8x .
T	Cycles traffic patterns (Balanced, Rush Hour, Heavy N-S, Heavy E-W, Light).
R	Resets simulation with a fresh synchronized random seed.
ESC / Q	Exits the simulation cleanly.

6. Streamlit Analytics Dashboard Structure

Navigate the browser interface at <http://localhost:8501> to explore each dedicated section:

- **1_Overview:** Problem statement, Markov Decision Process (MDP) state-action formalization, and reward architecture.
- **2_Training:** Interactive Plotly charts of episode reward convergence, rolling average wait time, and epsilon decay curves.
- **3_Benchmark:** Head-to-head empirical comparison (Wait time: -28%, Throughput: +4.1%, Emergency: -81.4%).
- **4_Emergency:** Preemption logic explanation and conflict resolution workflow.
- **5_Dataset:** Kaggle Urban Traffic Light Control dataset calibration proving simulation fidelity against real-world traffic data.

7. Frequently Encountered Issues & Troubleshooting

Issue Observed	Root Cause & Instant Resolution
'python' is not recognized	Python was not added to system PATH during install. Re-run installer, select 'Modify', check 'Add Python to environment variables', and restart terminal.
Port 8501 is in use	Another Streamlit instance is running. Kill existing process or launch on alternate port: <code>streamlit run streamlit_app/app.py --server.port 8502</code> .
Pygame DLL error (App Control)	Certain Windows Defender Application Control policies block standard Pygame DLLs. Our requirements.txt uses pygame-ce which resolves this.
Permission denied (PowerShell script)	PowerShell restricts unsigned scripts by default. Run: <code>Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass</code> before activating the venv.
ModuleNotFoundError: No module...	Virtual environment was not activated or dependencies were not installed. Ensure virtual environment is active ((venv) prefix in prompt) and run <code>pip install -r requirements.txt</code> .

End of Setup Manual. For issues, code updates, or star support, visit: <https://github.com/SauravLamba17/RL-Smart-Traffic-Light>